Notes: Banerjee and Duflo (RES, 2014)

Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program

Contents

1 Big picture

1.1 Question

The paper asks whether relatively large firms in a developing economy want to borrow more than they can obtain. The object of interest is not simply whether firms accept cheap bank credit. Any firm may accept subsidized or preferential credit if it is cheaper than outside credit. The question is whether additional directed credit relaxes a true financing constraint and expands production.

The empirical setting is a directed lending program in India. In 1998, the definition of the small-scale industry priority sector was expanded from firms with plant and machinery investment below Rs. 6.5 million to firms below Rs. 30 million. In 2000, the reform was partially reversed: firms with investment above Rs. 10 million were removed from priority-sector eligibility.

1.2 Core answer

The authors find strong evidence that many of these firms were credit constrained.

- When firms became eligible for priority-sector lending, their bank credit limits rose relative to smaller firms.
- When many of them lost eligibility, their bank credit limits fell relative to firms that remained eligible.
- Sales, costs, and profits moved in the same direction as credit.
- Interest rates did not move differentially, so the result is not simply a price response.
- There is little evidence that directed credit substituted for other credit.

Interpretation: The extra bank credit was used to finance more production, not merely to replace market borrowing. This is the empirical signature of a binding credit constraint.

1.3 Main takeaways

- The paper distinguishes **credit rationing at one lender** from a true **aggregate credit constraint**.
- The 1998 expansion and 2000 reversal generate two policy shocks with opposite signs, strengthening identification.
- Instrumental-variable estimates imply very high returns to working-capital loans.
- The implied marginal return before interest is about 105.5%, far above bank deposit returns and available macro estimates of marginal products of capital.
- The findings suggest severe capital misallocation: high-return firms do not receive enough credit.

2 Incremental contribution of the paper

2.1 What the paper adds relative to prior work on credit constraints

Earlier micro evidence on credit constraints in developing countries mainly came from farms, microenterprises, or very small firms. This paper studies registered, medium-sized Indian firms that are large enough to matter for aggregate capital allocation.

Incremental point: Credit constraints are not only a microenterprise problem. They can affect larger manufacturing firms with substantial production capacity.

2.2 What the paper adds methodologically

The paper uses two regulatory changes as quasi-experiments:

- an expansion of eligibility in 1998;
- a partial contraction of eligibility in 2000.

This creates a useful design because the same class of firms is treated positively and then partially untreated. The authors can test whether the signs reverse and whether implied estimates line up across the two reforms.

Incremental point: The paper does not rely only on investment-cash flow correlations. It uses policy-induced variation in credit supply and checks whether real activity responds.

2.3 What the paper adds conceptually

The paper clarifies the difference between accepting cheap credit and being credit constrained. An unconstrained firm might take all the cheap directed credit it can get, but it should mostly replace more expensive market borrowing. A constrained firm should expand total borrowing, output, and profits.

Incremental point: The key test is not whether firms borrow more from the bank. The key test is whether real outcomes rise without full substitution away from other borrowing.

2.4 What the paper adds for misallocation

The estimated marginal return to additional working-capital credit is far above plausible outside interest rates and macro-level estimates of the marginal product of capital. This is direct evidence that capital is not flowing to some high-return uses.

Practical implication: Improving bank lending technology, collateral enforcement, and loan-officer incentives could have large productivity effects.

3 Institutions and data

3.1 Indian banking and priority-sector lending

India's banking sector in this period was dominated by public-sector banks. Banks were required to lend at least 40% of net credit to the priority sector. If they failed to satisfy this target, they faced penalties or had to place funds with government agencies at low returns.

The relevant category is small-scale industry lending. Before the 1998 reform, firms qualified if plant and machinery investment was below Rs. 6.5 million. The 1998 reform expanded eligibility up to Rs. 30 million. The 2000 reversal removed firms above Rs. 10 million from eligibility.

Interpretation: The policy did not simply lower a market-wide interest rate. It changed which firms counted toward a binding regulatory lending target.

3.2 Working capital credit

The bank mostly provides short-term working-capital credit through credit lines. The credit line has a pre-specified limit and interest rate. Firms draw on the line when needed.

The relevant outcome is therefore the **working capital limit**. This is the amount of bank credit available to the firm at a point in time, not necessarily a long-term capital investment loan.

3.3 Data construction

The authors collected firm-level information from loan folders at a large public-sector bank. The sample contains firms that were clients of the bank and had plant and machinery investment below Rs. 30 million as of the relevant period. The data include:

- credit limits and interest rates;
- balance sheet and profit and loss information;
- sales, costs, profits, and bank loan information;
- firm characteristics used for heterogeneous effects.

The sample is not designed to estimate the aggregate effect of priority-sector lending on the whole economy. Instead, it uses the policy as an exogenous source of differential credit growth across firm classes.

4 Conceptual framework

4.1 Credit rationing versus credit constraints

A firm may be **rationed** at a cheap source of credit without being truly credit constrained. Suppose the firm can borrow a limited amount at a subsidized bank rate r_b and unlimited additional credit at a market rate r_m. If $r_b < r_m$, the firm wants all subsidized credit it can obtain, even if it is not constrained overall.

The firm is truly **credit constrained** if total available capital is below the level it would choose at the highest rate it can actually access.

4.2 Production and working capital

The firm pays a sunk cost and then produces with inputs. Some inputs must be financed out of working capital. The working-capital constraint is:

$$I_1 p_1 + I_2 p_2 + \cdots + I_m p_m \leq k_b + k_m = k, \tag{1}$$

where k_b is bank capital, k_m is market capital, and k is total working capital.

The reduced-form production relation is written as:

$$R = f_t(k_t), \qquad f_t'(k_t) > 0. \tag{2}$$

4.3 Prediction 1: unconstrained firms

If firms can borrow as much as they want at the market rate, expanding subsidized bank credit should mainly reduce market borrowing. Output need not change unless the cheap bank credit fully replaces all market credit and pushes the firm to a lower marginal cost of capital.

Empirical implication: If newly eligible firms are unconstrained, directed bank credit should show up mostly as substitution, not as higher sales and profits.

4.4 Prediction 2: constrained firms

If firms are constrained in total borrowing, an expansion of bank credit raises total working capital. Sales, costs, and profits should rise, and market borrowing should not fall one-for-one.

Empirical implication: Real activity should move with the credit shock. This is exactly what the paper tests.

4.5 Bank lending incentives and evergreening

The paper also develops a simple supply-side model of bank officers. Loan officers are rewarded for lending but penalized for defaults. This can generate two patterns:

- high-quality firms may receive more credit because they are safer;
- weak firms may also receive credit to prevent default, a form of evergreening.

This helps explain why ordinary cross-sectional correlations between loans and outcomes may be misleading. The policy shock isolates the marginal firms whose credit changes because of the reform.

5 Empirical design

5.1 Treatment groups

The paper uses three groups:

- **Small firms**: plant and machinery investment below Rs. 6.5 million; already eligible before 1998.
- **Medium firms**: investment between Rs. 6.5 million and Rs. 10 million; newly eligible in 1998 and remain eligible after 2000.
- **Biggest firms**: investment between Rs. 10 million and Rs. 30 million; newly eligible in 1998 and lose eligibility in 2000.

The contrast across these groups and periods identifies credit-supply variation.

5.2 First-stage credit regression

The basic credit-limit regression for the 1998 expansion is:

$$\Delta \log k_{it}^{b} = \alpha + \beta POST_t + \gamma BIG_i + \delta(BIG_i \times POST_t) + \varepsilon_{it}. \tag{3}$$

For the 2000 reversal, the analogous specification uses $BIGGEST_i \times POST2_t$. The pooled specification separates medium and biggest firms and includes interactions for both the expansion and contraction periods.

5.3 Reduced-form outcome regressions

The same interaction terms are used to study outcomes:

$$\Delta y_{it} = \alpha + \beta POST_t + \gamma BIG_i + \delta(BIG_i \times POST_t) + u_{it}, \tag{4}$$

where y can be sales, costs, profits, interest rates, or credit utilization.

5.4 Instrumental-variable estimate

The interactions between eligibility status and reform periods instrument for changes in bank working-capital limits. The IV estimate answers:

> How much do sales, costs, or profits change when bank credit changes because of priority-sector eligibility?

The main identification assumption is that treated and comparison firms would not have had different outcome trends absent the credit changes. The paper strengthens this assumption by using two reforms with opposite signs and by testing overidentifying restrictions.

6 Main findings

6.1 Credit responds to the policy shocks

The 1998 expansion increased credit growth for newly eligible larger firms relative to small firms. The 2000 reversal reduced credit growth for the firms that lost eligibility. The effects operate mostly through the size of credit-limit changes, not through the probability that the limit is revised.

Interpretation: Loan officers do not appear to suddenly open new files for many firms. Instead, they adjust the intensive margin of credit granted to firms already being reviewed.

6.2 Interest rates do not explain the results

The authors find no meaningful differential change in interest rates around the reforms. This matters because if the credit response were only a price response, we would not know whether quantities reflect rationing. Instead, firms receive different credit limits at similar interest rates.

Interpretation: Firms are rationed with respect to bank credit at the offered rate.

6.3 Sales, costs, and profits move with credit

Firms that gain credit expand sales and costs; firms that lose credit contract. The effect is concentrated in firms whose limits actually change. Firms without limit changes do not show the same outcome movements.

This supports the claim that credit affects production rather than simply tracking unrelated firm-specific trends.

6.4 No evidence of full substitution

The authors check whether firms are merely replacing other borrowing with bank borrowing. The sales response remains when they restrict to firms that still have non-bank liabilities. This is inconsistent with the pure substitution story.

Interpretation: The extra directed credit relaxes a real financing constraint.

6.5 Estimated returns are very high

The IV estimates imply:

- elasticity of sales with respect to bank credit: about 0.75;
- elasticity of costs with respect to bank credit: about 0.70;
- a Rs. 100,000 increase in loans raises sales by roughly Rs. 829,000 and costs by roughly Rs. 739,000 for the average firm;
- net profit rises by roughly Rs. 89,500 after repaying interest;
- adding back the priority-sector interest rate of about 16%, the implied marginal return before interest is about 105.5%.

Interpretation: These returns are too high to be reconciled with unconstrained access to credit at plausible market interest rates.

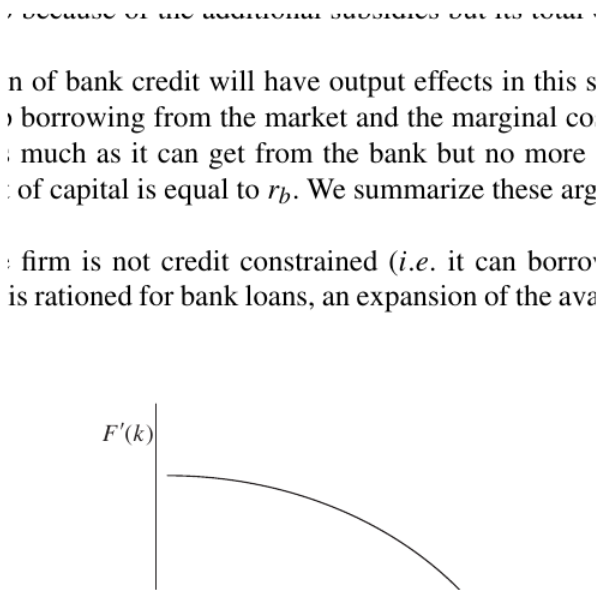
6.6 Misallocation and local increasing returns

Depositors earned about 10% or less in safe bank accounts, while the marginal return to working-capital loans for treated firms is estimated to exceed 100%. Macro estimates of marginal products of capital in India or comparable economies are far lower.

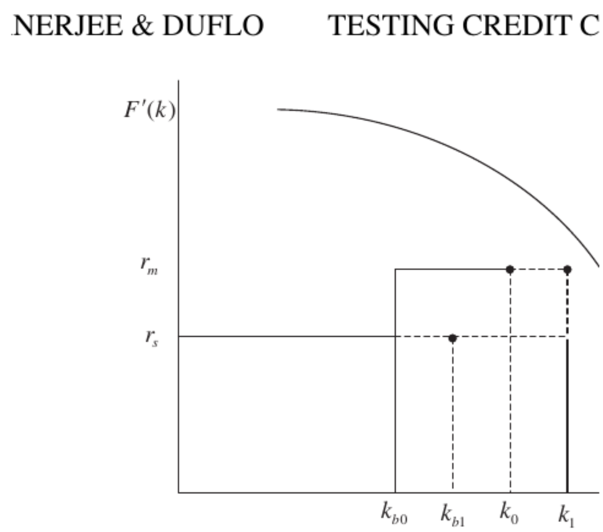
The production-function exercise also suggests a lower bound above one for local returns to scale in working capital. If an unconstrained firm faced locally increasing returns, it would not stop at an interior borrowing level. Therefore, at least some firms must be credit constrained.

7 Tables and figures from the paper

7.1 Theory figures: Figure 1: unconstrained substitution case and Figure 2: constrained expansion case



(a) Figure 1: unconstrained substitution case



(b) Figure 2: constrained expansion case

Read:

- Figure 1 shows the case where market credit is available elastically at r_m. Extra cheap bank credit first substitutes for market borrowing, leaving total capital unchanged.
- Figure 2 shows the credit-constrained case. Extra bank credit raises total capital from k_0 to k_1 because the firm cannot freely borrow the desired amount from the market.

Economic message: These two figures are the conceptual core of the paper. Both constrained and unconstrained firms may want subsidized bank credit. The difference is that only constrained firms use it to expand real activity.

7.2 Data and lending environment: Table 1: summary statistics and Table 2: characteristics of loans

TABLE 1

Descriptive statistics

	Levels		Change(t)-(t-1)	
	Entire sample (1)	Change in loans not missing (2)	Entire sample (3)	Change in loans not missing (4)
Panel A: Loans and interest rates				
Working capital Loan limit (this bank)	87.66 (237.04) 1226	96.29 (258.2) 928	10.29 (59.92) 966	7.46 (55.32) 928
log(working capital loan limit) (this bank)	3.39 (1.47) 1208	3.44 (1.5) 928	0.07 (0.24) 928	0.07 (0.24) 928
Working capital Loans limit (all banks)	87 (246) 1102	97 (273) 807	10 (69) 842	7 (67) 807
log(working capital loans limits) (all banks)	3.36 (1.48) 1085	3.41 (1.51) 807	0.06 (0.26) 807	0.06 (0.26) 807
Other bank loans Positive	0.0120 (0.11) 1748	0.004 (0.06) 807	0.0000 (0.14) 1748	−0.007 (0.1) 807
Other bank loans (level)	1.65 (25.86) 1748	2.23 (36.54) 807	0.00 (22.54) 1748	−0.62 (30.9) 807
Interest rate	15.75 (1.63) 1142	15.58 (1.59) 896	−0.32 (0.94) 876	−0.32 (0.94) 856
log(interest rate)	2.75 (0.18) 1142	2.74 (0.19) 896	−0.02 (0.16) 878	−0.02 (0.17) 858
Panel B: Credit utilization and firm performance				
Account reaches the limit	0.72 (0.45) 522	0.69 (0.46) 380	−0.01 (0.44) 247	−0.01 (0.44) 233
log(account turnover/ granted limit)	2.15 (0.95) 384	2.15 (0.96) 308	0.09 (0.72) 170	0.11 (0.71) 167
Sales	709.33 (2487.24) 1259	820.70 (2714.88) 746	108.64 (653.62) 1041	86.66 (598.64) 739
log(sales)	5.49 (1.44) 1248	5.64 (1.46) 740	0.17 (0.53) 1029	0.09 (0.45) 732
Net profit	36.53 (214.11) 1259	42.49 (237.16) 747	6.08 (61.33) 1043	4.00 (58.32) 741
log(costs)	5.45 (1.45) 1245	5.61 (1.45) 739	0.16 (0.51) 1022	0.09 (0.43) 729

Notes: Columns 1 and 2 present the mean level of each variable, with the standard error in parentheses and the number of observations on the third line. Columns 3 and 4 present the mean change in each variable, with the standard error in parentheses and the number of observations on the third line. All values are expressed in current Rs.10,000. The account turnover is the sum of all the payments that have been made using the firm's credit line over the year.

(a) Table 1: summary statistics

rtions of cases in which						
ed limit remained the same	0.66	0.64	0.65	0.76	0.73	0.73
was attained by the borrower	0.81	0.67	0.77	0.76	0.68	0.57
ed limit from banking system remained the same	0.66	0.63	0.63	0.76	0.73	n/a
num authorized limit has increased	0.63	0.74	0.73	0.58	0.77	0.74
ted sales have increased	0.72	0.67	0.73	0.71	0.70	0.71
ed limit <maximum authorized limit	0.60	0.63	0.60	0.50	0.47	0.22
ed limit <0.20*projected sales	0.85	0.85	0.79	0.82	0.82	0.81
s:						
granted limit/maximum authorized	0.88 (0.061)	0.81 (0.05)	0.90 (0.054)	0.83 (0.056)	0.99 (0.126)	1.00 (0.07)
granted limit/(0.20*predicted sales)	0.62 (0.041)	0.63 (0.037)	0.68 (0.034)	0.63 (0.055)	0.68 (0.064)	0.71 (0.062)
er of loans	175	208	205	175	163	124

· Each column present the data on the limit approved in a given year (to be used in the following year). Limits from banks were not collected in year 2002.

d the maximum authorized lending (a function of projected sales based on the bank's official ng policies, which the loan officer dutifully records on the file). The change in the credit that was actually sanctioned systematically fell short of what the bank itself determined to e firm's needs (which is also reported in the file). On average, the granted limit was 89% of

(b) Table 2: characteristics of loans

Read:

- Table 1 shows the core variables: working capital limits, interest rates, credit utilization, sales, costs, and profits.
- Table 2 shows that credit limits are sticky: in many years, the granted limit is unchanged even though many borrowers reach the limit.

Economic message: The environment is one in which credit quantities are administratively sticky and economically meaningful. The fact that borrowers often reach the credit limit makes the limit a plausible binding constraint.

7.3 Bank limit changes before the reforms: Table 3: changes in limits by firm characteristics and Table 4: average change in credit limit

		Propoportion of cases where		Mean of:	Proportion of cases where limit was changed	
	Proportion	Limit was increased	Limit was not changed	log(current limit) –log(past limit)	Client<=5 years	Client>5 years
	(1)	(2)	(3)	(4)	(5)	(6)
A-Has past utilization reached maximum?						
Yes	0.72	0.34	0.60	0.16	0.55	0.67
No	0.28	0.30	0.66	0.12	0.61	0.69
Difference		0.05	−0.05	0.03	−0.05	−0.02
		(0.054)	(0.056)	(0.04)	(0.081)	(0.059)
B-Have projected sales increased?						
Yes	0.71	0.43	0.52	0.19	0.54	0.56
No	0.29	0.25	0.61	0.06	0.50	0.73
Difference		0.18	−0.09	0.13	0.04	−0.17
		(0.076)	(0.079)	(0.053)	(0.114)	(0.083)
C-Have actual sales increased?						
Yes	0.71	0.33	0.62	0.13	0.61	0.68
No	0.29	0.25	0.69	0.12	0.70	0.74
Difference		0.08	−0.06	0.02	−0.09	−0.06
		(0.041)	(0.043)	(0.029)	(0.059)	(0.04)
D-Has profit over sale increased?						
Yes	0.56	0.29	0.67	0.11	0.64	0.71
No	0.44	0.35	0.61	0.16	0.61	0.70
Difference		−0.06	0.06	−0.05	0.03	0.01
		(0.042)	(0.044)	(0.028)	(0.059)	(0.043)
E-Has current ratio increased?						
Yes	0.53	0.32	0.62	0.12	0.61	0.70
No	0.47	0.29	0.67	0.14	0.67	0.72
Difference		0.03	−0.05	−0.02	−0.06	−0.03
		(0.038)	(0.04)	(0.027)	(0.052)	(0.039)

Notes: Each panel divides the sample in two subsamples, according to the answer to the question asked in the panel title. Column 1 gives the proportion of the sample that falls into each category. The first two rows in Columns (2) to (6) display the mean of the relevant variables in the subsample where the answer to the question in the panel title is yes (row 1 in each panel), and no (row 2 in each panel). Row 3 is the difference between row 1 and 3 in each panel. The standard errors are in parentheses in row 4.

3. THEORY: THE EFFECT OF SUBSIDIZED BANK CREDIT ON FIRM OUTCOMES

The goal of this section is to develop some intuition about how we expect firms to react to an increase in the supply of subsidized bank credit. We study the choice problem faced by a firm

(a) Table 3: changes in limits by firm characteristics

	Years		
	1996–1997	1998–1999	2000–2002
Firm's category			
A. Average change in limit			
Small	0.110	0.075	0.070
	(0.021)	(0.013)	(0.014)
Medium	0.040	0.093	0.011
	(0.032)	(0.030)	(0.025)
Biggest	0.093	0.147	0.000
	(0.064)	(0.040)	(0.031)
Firms with fixed capital	0.135	0.062	0.153
between Rs30 and 450 million (4)	(0.017)	(0.017)	(0.019)
B. Proportion of cases where limit was not changed			
Small	0.701	0.701	0.724
	(0.043)	(0.031)	(0.027)
Medium	0.667	0.608	0.798
	(0.088)	(0.055)	(0.040)
Biggest	0.625	0.692	0.769
	(0.183)	(0.075)	(0.053)
C. Average change in limit, conditional on change			
Small	0.366	0.252	0.253
	(0.045)	(0.035)	(0.045)
Medium	0.119	0.237	0.053
	(0.093)	(0.068)	(0.124)
Biggest	0.248	0.479	−0.002
	(0.137)	(0.062)	(0.138)
D. Fraction of cases where the limit was decreased			
Small	0.026	0.051	0.052
	(0.015)	(0.015)	(0.014)
Medium	0.067	0.076	0.087
	(0.046)	(0.03)	(0.028)
Biggest	0.000	0.000	0.123
			(0.041)

Notes: The first row of each panel presents the average of log(working capital limit granted at date t)-log(working capital limit granted at date t-1). Standard errors in parentheses below the average. "Small firms" are firms with investment in plant and machinery below Rs. 6.5 million. "Medium firms" are firms with investment in plant and machinery above Rs. 6.5 million and below Rs. 10 million. "Biggest firms" are firms with investment in plant and machinery above Rs. 10 million and below Rs. 30 million Firms with fixed capital between Rs. 30 million and Rs. 450 million: PROWESS data set, Center for Monitoring the Indian Economy.

Since the granted limit, as well as all the outcomes we will consider, are very strongly auto-correlated, we focus on the proportional change in this limit, *i.e.* log(limit granted in year t)−log(limit granted in year t-1).[9] As motivation, Table 4 shows the average change in the credit limit faced by the firm in the three periods of interest (loans granted before the change in January 1998, between January 1998 and January 2000, and after January 2000) separately for the largest firms (investment in plant and machinery measured before 1998 between Rs. 10 million and Rs. 30 million), the medium-sized firms (investment in plant and machinery measured before 1998 between Rs. 6.5 and Rs. 10 million), and the smaller firms (investment in plant and machinery measured before 1998 below Rs. 6.5 million).

(b) Table 4: average change in credit limit

Read:

- Table 3 shows that routine credit-limit changes are not strongly explained by simple indicators of need, such as whether the firm hit the limit or whether profits rose.
- Table 4 shows the policy timing: larger firms receive faster credit-limit growth after the 1998 expansion and much slower growth after the 2000 reversal.

Economic message: Ordinary bank lending rules are rigid, but the policy shocks move credit in the predicted direction. This makes the reforms useful instruments for credit supply.

7.4 First stage and interest rates: Table 5: effect of priority-sector reforms on credit and Table 6: effects on interest rates and utilization

TABLE 5

Effect of the priority sector reforms on credit (OLS regressions)

	Dummy equal to 1 if			Log(working capital limit availble at t)-log(working capital limit available at t-1)			
	Limit was changed between t and t-1	Limit increased between t and t-1	Limit decreased between t and t-1	Whole sample	Sample with change in limit	Whole sample	Sample with change in limit
						Sales information not missing	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PANEL A: t = 1997–2000							
Post	0.000	−0.026	0.026	−0.034	−0.115	−0.025	−0.102
	(0.050)	(0.052)	(0.024)	(0.026)	(0.074)	(0.028)	(0.071)
Big	−0.043	0.016	0.027	−0.059	−0.218	−0.055	−0.206
	(0.052)	(0.051)	(0.041)	(0.028)	(0.088)	(0.028)	(0.082)
Post*big	−0.022	0.050	−0.028	0.095	0.271	0.087	0.259
	(0.087)	(0.079)	(0.044)	(0.033)	(0.102)	(0.032)	(0.099)
	487	487	487	487	155	453	152
PANEL B: t = 1999–2003							
Post2	0.069	−0.073	0.004	−0.027	−0.038	−0.028	0.001
	(0.032)	(0.037)	(0.024)	(0.024)	(0.075)	(0.026)	(0.077)
Biggest	0.017	0.041	−0.058	0.067	0.232	0.057	0.251
	(0.129)	(0.131)	(0.017)	(0.059)	(0.063)	(0.058)	(0.057)
Post2*biggest	0.008	−0.127	0.119	−0.121	−0.442	−0.128	−0.549
	(0.179)	(0.172)	(0.033)	(0.082)	(0.191)	(0.080)	(0.171)
	769	769	769	769	217	569	168
PANEL C: t = 1997–2003							
Post*biggest (γ_{3kb})	0.067	−0.041	−0.026	0.089	0.346	0.076	0.352
	(0.150)	(0.150)	(0.024)	(0.059)	(0.146)	(0.059)	(0.145)
Post*medium (γ_{4kb})	−0.059	0.076	−0.016	0.088	0.233	0.083	0.221
	(0.098)	(0.090)	(0.051)	(0.041)	(0.122)	(0.042)	(0.119)
Post2*biggest (γ_{5kb})	0.054	−0.176	0.122	−0.142	−0.482	−0.150	−0.581
	(0.175)	(0.170)	(0.033)	(0.077)	(0.181)	(0.076)	(0.157)
Post2*medium (γ_{6kb})	0.168	−0.177	0.010	−0.077	−0.185	−0.078	−0.170
	(0.034)	(0.052)	(0.040)	(0.044)	(0.167)	(0.040)	(0.159)
	924	924	924	924	265	718	215

Notes: Each panel is a separate regression. Each column presents a regression of column heading on the variables listed in each panel. For consistency of notation across Tables 5 to 9, we display credit available in year t (granted in year t-1). The dummy "post" is equal to 1 for credit available in 1999 and 2000 (granted in year 1998 and 1999), zero othertwise. The dummy "post2" is equal to 1 for credit available in 2001-2002-2003 (granted in years 2000, 2001, and 2002), zero otherwise. The dummy "big" is equal to 1 for firms with investment in plant and machinery larger than Rs. 6.5 millions, zero otherwise. The dummy "medium" is equal to 1 for firms with investment in plant and machinery between Rs. 6.5 and Rs. 10 million. The dummy "biggest" is equal to 1 for firms with investment in plant and machinery larger than Rs. 10 million. In addition to the coefficients displayed, the regression in panel C includes the dummies "post", "post2", "medium", "biggest". Standard errors (corrected for clustering at the sector level) are in parentheses below the coefficient.

(a) Table 5: effect of priority-sector reforms on credit

TABLE 6

Effect of the reforms on interest rate and limit utilization (OLS regressions)

	Complete sample				Sample where limit was changed		
	Interest rate$_t$ − interest rate$_{t-1}$	log(interest rate)$_t$ −log(interest rate)$_{t-1}$	Dummy for interest rate decline	log(turnover/limit)$_t$ −log(turnover/limit)$_{t-1}$	Interest rate$_t$ − interest rate$_{t-1}$	log(interest rate)$_t$ −log(interest rate)$_{t-1}$	Dummy for interest rate decline
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. t = 1997–2000							
Post	−0.165	−0.010	0.280	0.154	−0.127	−0.007	0.279
	(0.128)	(0.008)	(0.074)	(0.174)	(0.249)	(0.015)	(0.151)
Big	−0.002	0.000	0.098	0.412	−0.036	−0.002	0.052
	(0.132)	(0.008)	(0.106)	(0.188)	(0.241)	(0.014)	(0.153)
Post*big	0.073	0.002	−0.135	−0.145	0.163	0.009	−0.144
	(0.169)	(0.011)	(0.125)	(0.249)	(0.337)	(0.020)	(0.225)
	430	430	430	93	141	141	141
B. t = 1999–2002							
Post2	0.035	−0.009	−0.029	0.050	−0.146	−0.008	0.225
	(0.072)	(0.013)	(0.038)	(0.102)	(0.167)	(0.013)	(0.068)
Biggest	−0.062	−0.007	−0.010	0.986	−0.077	−0.004	0.039
	(0.110)	(0.008)	(0.063)	(0.580)	(0.188)	(0.011)	(0.140)
Post2*biggest	0.099	0.020	0.001	−0.872	0.206	0.013	−0.036
	(0.147)	(0.017)	(0.098)	(0.871)	(0.385)	(0.026)	(0.184)
	719	721	721	139	203	203	203

Notes: Each panel is a separate regression. Each column presents a regression of column heading on the variables listed in each panel. The interest rate is the interest rate on credit used at date t (granted at date t-1). The dummy "post" is equal to 1 for year 1999 and 2000 (limit granted in years 1998 and 1999), zero otherwise. The dummy "post2" is equal to 1 for years 2001-2002-2003 (limit and interest rate granted in years 2000, 2001, and 2002), zero otherwise. The dummy "big" is equal to 1 for firms with investment in plant and machinery larger than Rs. 6.5 million, zero otherwise. The dummy "medium" is equal to 1 for firms with investment in plant and machinery between Rs. 6.5 and Rs. 10 million. The dummy "biggest" is equal to 1 for firms with investment in plant and machinery larger than Rs. 10 million. Standard errors (corrected for clustering at the sector level) are in parentheses below the coefficient.

(b) Table 6: effects on interest rates and utilization

Read:

- Table 5 is the first stage. The 1998 expansion raises credit limits for newly eligible firms; the 2000 reversal cuts credit for firms that lose eligibility.
- Table 6 shows little evidence of differential interest-rate changes. The main change is quantity access, not price.

Economic message: The reforms affect bank credit supply. Because interest rates do not adjust in parallel, the credit-limit changes are evidence of rationing at the bank rather than a standard demand response to lower prices.

7.5 Real effects and overidentification: Table 7: effects on sales, costs, and profits and Table 8: overidentification tests

	Dependent variables				
	Log(sales)_{t+1}-log(sales)_t			Log(cost)_{t+1}	Log(profit)_{t+1}
	Complete sample	Sample without substitution	$\text{log(sales/loans)}_{t+1}$ $\text{log(sales/loans)}_t$	-log(cost)_t	-log(profit)_t
	(1)	(2)	(3)	(4)	(5)
A. 1996–1999					
1. Sample with changes in limit					
Post	0.013	0.000	0.124	0.035	0.063
	(0.085)	(0.095)	(0.101)	(0.101)	(0.145)
Post*big	0.194	0.168	−0.126	0.187	0.538
	(0.106)	(0.118)	(0.094)	(0.097)	(0.281)
	152	136	152	151	141
2. Sample without change in limit					
Post	0.012	0.023	0.031	0.022	−0.316
	(0.052)	(0.045)	(0.048)	(0.049)	(0.177)
Post*big	0.007	0.022	0.016	0.004	0.280
	(0.074)	(0.081)	(0.065)	(0.064)	(0.473)
	301	285	301	301	250
3. Whole sample					
Post*big	0.071	0.071	−0.032	0.067	0.316
	(0.068)	(0.069)	(0.065)	(0.054)	(0.368)
	453	421	453	452	391
B. 1998–2001					
1 Sample with changes in limit					
Post2	−0.041	−0.041	−0.039	−0.072	0.030
	(0.088)	(0.087)	(0.121)	(0.088)	(0.263)
Post2*biggest	−0.403	−0.387	0.143	−0.374	−0.923
	(0.207)	(0.196)	(0.206)	(0.279)	(0.639)
	168	150	169	168	151
2. Sample without change in limit					
Post2	−0.012	−0.044	−0.033	−0.047	0.042
	(0.043)	(0.043)	(0.039)	(0.049)	(0.124)
Post2*biggest	−0.092	−0.045	−0.101	−0.047	0.170
	(0.108)	(0.128)	(0.088)	(0.086)	(0.56)
	401	380	401	399	321
3. Whole sample					
Post*big	−0.143	−0.113	−0.023	−0.100	−0.253
	(0.111)	(0.134)	(0.153)	(0.093)	(0.496)
	569	530	570	567	472

Notes: Each panel is a separate regression. Each column presents a regression of column heading on the variables listed in each panel. The dummy "post" is equal to 1 in years 1999 and 2000, zero otherwise. The dummy "post2" is equal to 1 in years 2001–2002, zero otherwise. The dummy "big" is equal to 1 for firms with investment in plant and machinery larger than Rs. 6.5 million, zero otherwise. The dummy "biggest" is equal to 1 for firms with investment in plant and machinery larger than Rs. 10 million. Standard errors (corrected for clustering at the sector level) are in parentheses below the coefficient. In addition to coefficient displayed, the regressions in panels A1-A3 include a dummy for big. In addition to coefficient displayed, the regressions in panels B1-B3 include a dummy for biggest.

To avoid this problem, we look at the direct impact of the reform on the logarithm of cost (defined as sales-profits), which is always defined. The effect on profit for any particular firm or for the average firm can then be recovered from the estimate of the reform on sales and costs, without sample selection bias. The increase in sales is accompanied by an increase in cost of comparable magnitude: the coefficient on the $BIG*POST$ interaction is 0.187 in the sample with change in limit, and only 0.005 in the sample without change in limit.

(a) Table 7: effects on sales, costs, and profits

	Dependent variables		
	Log(sales)_t-log(sales)_{t-1}		Log(costs)_t
	Complete sample	Sample without substitution	-log(cost)_{t-1}
	(1)	(2)	(3)
Post*biggest (γ_{3y})	0.238	0.235	0.205
	(0.153)	(0.162)	(0.151)
Post*medium (γ_{4y})	0.182	0.146	0.183
	(0.121)	(0.134)	(0.109)
Post2*biggest (γ_{5y})	−0.421	−0.400	−0.384
	(0.197)	(0.186)	(0.279)
Post2*med (γ_{6y})	−0.091	−0.095	−0.072
	(0.113)	(0.115)	(0.112)
	215	193	215
Ratio 1: γ_{3y}/γ_{3kb}	0.676	0.666	0.583
Ratio 2: γ_{4y}/γ_{4kb}	0.825	0.662	0.829
Ratio 3: γ_{5y}/γ_{5kb}	0.725	0.689	0.660
Ratio 4: γ_{6y}/γ_{6kb}	0.535	0.561	0.424
Test ratio 1=ratio2	0.06	0.00	0.17
(p value)	(0.81)	(0.99)	(0.68)
Test ratio 1=ratio2=ratio3	0.07	0.00	0.22
(p value)	(0.97)	(1.00)	(0.90)
Test ratio 1=ratio2=ratio3=ratio4	0.24	0.02	0.59
(p value)	(0.97)	(1.00)	(0.90)

Notes: All the regressions are estimated in the sample were the limit was changed. The dummy "post" is equal to 1 in years 1999 and 2000, zero otherwise. The dummy "post2" is equal to 1 in years 2001–2002, zero otherwise. The dummy "big" is equal to 1 for firms with investment in plant and machinery larger than Rs. 6.5 millions, zero otherwise. The dummy "biggest" is equal to 1 for firms with investment in plant and machinery larger than Rs. 10 million. Standard errors (corrected for clustering at the sector level) are in parentheses below the coefficient. In addition the the coefficients displayed, the regressions include dummies for post, post2, medium, and biggest. The parameters in parenthesis refer to equation (6) in the text. The ratios are computed using the parameters of equation (5) in the text, displayed in column 7 and panel C of table 5.

strategies pursued by the large firms. In order to answer this question, we use data on Non Performing Assets (NPAs). Since it takes at least a year for a loan that has gone bad to be officially qualified as an NPA, we treat the years 1998 and 1999 as the "pre" period, the year 2000 and 2001 as the period following the expansion, and 2002 as the period following the contraction. In 1998 and 1999, 1% of the loans to medium and large firms, and 5% of the loans to small firms,

(b) Table 8: overidentification tests

Read:

- Table 7 shows that sales and costs rise when credit expands and fall when credit contracts, especially among firms with actual limit changes.
- Table 8 pools the experiments and reports ratios implied by different instruments; the overidentification tests do not reject equality of the implied effects.

Economic message: The real response mirrors the credit response. The same economic effect appears across the expansion and contraction episodes, which strengthens the causal interpretation.

7.6 IV estimates and heterogeneous effects: Table 9: IV and OLS estimates and Table 10: which firms are most affected

TABLE 9

Effect of working capital loans on sales and profit, IV and OLS estimates

	Dependent variables					
	2SLS	2SLS	2SLS	2SLS	2SLS	OLS
	Sample with change	Sample with change	Sample with change	Sample with change	Complete sample	Complete sample
	1997–2000	1999–2002	1997–2002	1997–2002	1997–2002	1997–2002
				No SSI products		
Regressor:	(1)	(2)	(3)	(4)	(5)	(6)
A. $\log(\text{sales}_t)$-$\log(\text{sales}_{t-1})$						
log(working capital limit_t)	0.75	0.73	0.76	0.50	0.93	0.20
-log(working capital limit_t-1)	(0.37)	(0.35)	(0.32)	(0.35)	(1.12)	(0.09)
observations	152	168	215	190	718	718
B. $\log(\text{cost}_t)$-$\log(\text{cost}_{t-1})$						
log(working capital limit_t)	0.72	0.68	0.70	0.44	0.67	0.23
-log(working capital limit_t-1)	(0.36)	(0.44)	(0.4)	(0.5)	(0.82)	(0.08)
observations	151	168	215	189	716	716
C. $\log(\text{profit}_t)$-$\log(\text{profit}_{t-1})$						
log(working capital limit_t)	1.79	1.89	2.00	2.02	2.08	0.15
-log(working capital limit_t-1)	(0.94)	(1.49)	(0.996)	(0.99)	(3.26)	(0.19)
observations	141	151	192	166	598	598

Notes: Standard errors (corrected for clustering at the sector level and heteroskedasticity) in parentheses below the coefficients. Each panel and each column present the result for a separate regression. The regressions in column 1 control for the "post" and "big" dummy (defined as in previous tables) and use the interaction big*post as instrument. The regressions in column 2 control for the "post2" and "biggest" dummy (defined as in previous tables) and use the interaction biggest*post2 as instrument. The regressions in columns 3, 4 and 5 control for the dummies "post", "post2", "big" and "biggest" (defined as before) and use the interactions "post*med" "post2*biggest" and "post*biggest" as instruments.

(a) Table 9: IV and OLS estimates

	Sample cuts			
	By ownership		By ratio number of employees/investment in plant and machinery	
	Privately owned	Partnerships or Ltd companies	Below median	Above median or missing
	(1)	(2)	(3)	(4)
Panel A: First stage—Dependent variable: log(working capital limit_t)-log(working capital limit_t-1)				
Big* post (sample with change in limit)	0.40	0.090	0.65	0.15
	(0.13)	(0.06)	(0.3)	(0.12)
Panel B: Reduced form—Dependent variable: log(salest+1)-log(salest)				
B.1 sample with change in limits				
big*post	0.32	0.029	0.73	0.16
	(0.13)	(0.22)	(0.35)	(0.12)
B.2 sample without change in limit				
big*post	−0.078	0.155	−0.18	−0.09
	(0.13)	(0.12)	(0.26)	(0.06)
B.3 overall sample				
big*post	0.070	0.084	0.58	0.00
	(0.097)	(0.087)	(0.38)	(0.05)
Panel C: IV—Effect of loans on profit (estimated in the sample with change in limit)				
log(working capital limit_t)	0.79	0.478	1.13	1.19
-log(working capital limit_t-1)	(0.39)	(3.81)	(0.31)	(1.31)
-log(working capital limit_t-1)				

(b) Table 10: which firms are most affected

Read:

- Table 9 reports the key IV estimates: sales elasticity around 0.75, cost elasticity around 0.70, and a much smaller OLS estimate, consistent with selection in ordinary lending.
- Table 10 shows stronger responses for privately owned firms and for firms with lower employee-to-plant-and-machinery ratios.

Economic message: The marginal effect of policy-induced credit is large. The heterogeneity is consistent with the idea that firms with fewer alternative financing sources or more scope to expand labor-intensive production are most constrained.

8 Short summary

The paper uses India's priority-sector lending rules to test whether medium-sized firms are credit constrained. The key insight is that subsidized credit alone does not prove a constraint: unconstrained firms also want cheap credit. The decisive evidence is that policy-induced bank credit changes cause real output and profits to change, with no full substitution away from other borrowing.

The 1998 expansion and 2000 reversal provide two shocks with opposite signs. Credit limits, sales, costs, and profits all move in the predicted directions. Instrumental-variable estimates imply extremely high marginal returns to working-capital loans, suggesting that firms had profitable projects that they could not finance.

9 Conclusion

Banerjee and Duflo show that credit constraints can be severe even for relatively large firms in a developing economy. The paper's strongest contribution is its distinction between cheap-credit rationing and true credit constraints. By showing that directed credit expands production rather than merely replacing other credit, the paper provides direct evidence that bank credit supply affects real firm outcomes.

The broader lesson is about capital misallocation. If some firms can earn very high marginal returns but cannot obtain credit, then financial frictions are not just distributional or institutional details. They can directly reduce productivity and growth.